

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Examination, 2022-23

Paper No. CHB-401: Inorganic & Organic Chemistry-II

Time: 3 Hours

Maximum Marks: 70

Instructions: Answer five questions in total, including Question 1 from each section which is compulsory. Use separate answer books for Section A and Section B.

SECTION A

(Inorganic Chemistry-II – Marks: 35)

Question 1. Answer all parts of the following:

[5 × 2 = 10]

- (a) The water solubility of lithium halides is in the order $\text{LiBr} > \text{LiCl} > \text{LiF}$. However, this trend is reversed in the case of analogous silver halides. Explain.
- (b) What are the differences in characteristic properties of the blue colored solution and bronze colored phase obtained by addition of alkali metals in liquid ammonia?
- (c) What are the limitations of the Effective Atomic Number (EAN) rule?
- (d) Why do the 4d and 5d transition series elements show very similar ionic radii for corresponding elements? Explain.
- (e) Predict the relative coordination tendencies of F^- and I^- towards Fe^{3+} and Hg^{2+} using HSAB principles.

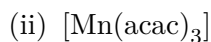
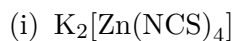
Question 2.

- (a) Give two examples each of protic and aprotic solvents and write autoionization reactions for them. [4]
- (b) Predict the product when $(\text{CH}_3)_3\text{N-PF}_2$ is reacted with: (i) BH_3 and (ii) BF_3 . Explain the reactivity difference between the two cases. [4]
- (c) State the physical properties of a solvent that determine its utility in chemical reactions. [4]

Question 3.

- (a) Write one example each of the following types of octahedral complexes. Draw the various stereoisomers and specify the enantiomeric pairs in each case: [4]
 - (i) $[\text{Ma}_2\text{b}_2\text{c}_2]$
 - (ii) $[\text{M}(\text{AA})_2\text{b}_2]$

(b) Write the IUPAC names of the following coordination compounds: [4]



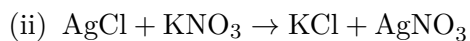
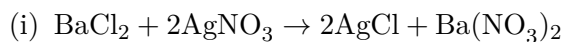
(c) Why is $[\text{Ni}(\text{en})_3]^{2+}$ more stable than $[\text{Ni}(\text{NH}_3)_6]^{2+}$ even though both are octahedral and nitrogen-donor complexes? Draw their structures. [4]

Question 4.

(a) Using Valence Bond Theory, predict the structure, hybridization, and spin-only magnetic moments of: $\text{Ni}(\text{CO})_4$, $[\text{NiCl}_4]^{2-}$, $[\text{Ni}(\text{CN})_4]^{2-}$, and $[\text{Ni}(\text{NH}_3)_6]^{2+}$. [6]

(b) Write the electronic configuration and number of unpaired electrons in the ground state of the following lanthanide ions: Ce^{3+} , Gd^{3+} , Eu^{3+} , and Tb^{3+} . [3]

(c) Predict the feasibility of the following reactions in water and liquid ammonia solvents along with reasons: [3]



SECTION B

(Organic Chemistry-III – Marks: 35)

Question 5. Answer all parts of the following: [5 × 2 = 10]

- (a) Write the chemical structures of α - and β -D-(+)-glucopyranose.
- (b) Discuss mutarotation of glucose in brief.
- (c) How do epimers and anomers differ? Explain with examples.
- (d) Give one method for the synthesis of Indigo dye from anthranilic acid.
- (e) Why do malachite green and fluorescein develop color in solution? Explain.

Question 6.

- (a) Discuss the orientation of electrophilic aromatic substitution reactions in pyridine. Explain why substitution occurs preferentially at position 3. [4]
- (b) Explain why pyridine *N*-oxide is more reactive than pyridine towards both electrophilic and nucleophilic substitution reactions. [4]
- (c) Write the products in the following reactions: [4]
 - (i) Quinoline $\xrightarrow{\text{NaNH}_2, 100^\circ\text{C}}$?
 - (ii) Pyrrole $\xrightarrow{\text{CH}_3\text{I}, \text{KOH}}$?

Question 7.

- (a) Discuss the synthesis of the following heterocyclic compounds with mechanisms: [6]
 - (i) Skraup quinoline synthesis
 - (ii) Bischler-Napieralski isoquinoline synthesis
- (b) Explain why anthracene is very reactive at the 9 and 10 positions. [3]
- (c) How will you establish the structure of indigotin? Explain its chemical degradation. [3]

Question 8.

- (a) How will you prepare the following compounds from D-(+)-glucose? Write stepwise mechanisms: [6]
 - (i) Fructose (Osazone method)
 - (ii) Mannose (Epimerization)

(b) Complete the following reactions:

[6]

